

decreases.

3

B

$\Delta p = \text{Area under } F\text{-}t \text{ graph}$

$$= (20 \times 4) + (0.5 \times 2 \times 20)$$

$$= 100 \text{ N s}$$

$$\Delta v = \Delta p / m = 50 \text{ m s}^{-1}$$

4

B

In Fig. A,